The unary relational interpretation of arithmetic R

October 20, 2026

Abstract

The library `Interpretation_of_R.relational` gives a fixed interpretation of the relational presentation of Tarski–Mostowski–Robinson arithmetic in the original rank-based pure type system. The public theorem `Theorem4.theorem4` combines realization of every axiom instance, preservation of an independently defined classical deduction system, proofs and internal refutations for all ground atoms, and a concrete failure of reflection for arbitrary sentences. The construction uses a unary Church carrier, negative equality tests, successor-closed observations, and uniformly certified finite selectors. This exposition follows the checked source declarations, including the fixed-rank invariant behind the quantified axiom schemes.

1 The theorem and its source language

The source theory, written $\widetilde{R}$, has zero, successor, equality, order, and ternary addition and multiplication relations A, M. Its terms are variables, zero, and iterated successor; addition and multiplication are predicates rather than source term constructors. For external natural numbers m, n, its axiom schemes are

$$\begin{aligned} A(\bar{m}, \bar{n}, x) &\leftrightarrow \overline{m+n} = x, \\ M(\bar{m}, \bar{n}, x) &\leftrightarrow \overline{mn} = x, \\ \bar{m} &\neq \bar{n} \quad (m \neq n), \\ x \leq \bar{n} &\leftrightarrow \bigvee_{i=0}^{n} x = \bar{i}, \\ x \leq \bar{n} &\vee \bar{n} \leq x. \end{aligned}$$

Free x is universally quantified. Both directions of the bounded-order biconditional are part of this relational presentation. `DirectRRelationalSyntax.v` defines the scoped source syntax and these five axiom constructors. Primitive logical syntax consists of falsity, implication, and universal quantification; negation, conjunction, biconditional, finite disjunction, and existential quantification are defined encodings. The inductive judgment `Relational.derives` has hypothesis and theory rules, implication and universal rules, classical double-negation elimination, equality reflexivity, and equality replacement. Its definition contains no target typing or inhabitation premise.

Write A^* for the fixed translation of a source sentence. The target judgment below is exactly `PTSrk.Typing.typ` at `Ranks.RankSorts`, using the original annotated `Syntax.Term nat n`.

Theorem 1.1 (The relational interpretation)**.** *The theorem `Theorem4.theorem4` packages the following four conclusions.*

1. *Every closed source axiom A has a closed original PTS inhabitant $p : A^*$.*
2. *If `Relational.derives [] A`, there is a closed original PTS inhabitant $p : A^*$.*
3. *Every true ground equality, comparison, addition atom, or multiplication atom has a proof of its translation; every false ground atom has an internal refutation of type $A^* \to \bot$.*

4. *Some source sentence has an inhabited translation but no derivation in the independently defined source calculus.*

This is the theorem labeled `main` in the harmonized manuscript. Its number is Theorem 4 there; it is distinct from the functional theorem and from the functional `Kernel.Theorem5Target` interface. The conjunction above has no assumed certificate package, all-axiom premise, or deduction preservation premise. The subsidiary files establish these facts before the final theorem assembles them.

2 The fixed carrier, domain, and logical translation

The PTS has sorts $U_i : U_0$, and products have rank $\max(a+1, b)$ when their domain and codomain have ranks a, b. There is no cumulativity or eta conversion. Define the unary Church carrier

$$D := \Pi X : U_1.X \to (X \to X) \to X : U_3, \qquad z := \lambda X\, x\, s.x, \qquad Sd := \lambda X\, x\, s.s(d\, X\, x\, s).$$

Displayed lambdas abbreviate fully annotated original terms. Let $\nu_n = S^n z$. The module `Unary` also defines two auxiliary operations

$$\begin{aligned} \mathrm{Add}(d, e) &:= \lambda X\, x\, s.d\, X\, (e\, X\, x\, s)\, s, \\ \mathrm{Mul}(d, e) &:= \lambda X\, x\, s.d\, X\, x\, (\lambda y : X.e\, X\, y\, s). \end{aligned}$$

Both have type $D \to D \to D$. They appear inside translated relations; they do not extend the source term syntax. The theorems `UnaryEvaluation.addition_numerals` and `multiplication_numerals` prove their ground beta equations. No equation identifying an arbitrary Church presentation with its eta expansion is needed.

The admitted domain is a constructor-closure certificate:

$$\mathrm{Good}(d) := \Pi Q : D \to U_{10}.Q(z) \to (\Pi u : D.Q(u) \to Q(Su)) \to Q(d) : U_{12}.$$

`Unary.good_zero_typed` and `good_successor_typed` certify its zero and successor constructors. `admitted_numeral_typed` produces the certificates for all external numeral indices. This definition supplies the precise bounded-rank induction needed later; it is not an induction rule for every raw D-inhabitant and does not assume the desired arithmetic axioms.

The fixed answer type is $\bot := \Pi Z : U_0.Z$, and $\top := U_0 \to U_0$ has its identity proof. Negation is $P \to \bot$, conjunction is $(P \to Q \to \bot) \to \bot$, and a finite disjunction is the flat continuation type

$$\bigvee_c (P_0, \ldots, P_k) := (P_0 \to \bot) \to \cdots \to (P_k \to \bot) \to \bot.$$

A proof type is stable when its double negation implies it. The translation preserves implication and interprets universal quantification by

$$(\forall x.B)^* := \Pi d : D.\, \mathrm{Good}(d) \to B^*(d).$$

Existential quantification follows from its source encoding and hence has a negative, guarded translation. `RelationalTranslation.formula_stable` proves stability for every translated formula. Classical elimination is therefore realized by PTS terms; no elimination from $\bot$ into arbitrary positive types is postulated.

3 Equality and order through observations

3.1 Negative equality tests

For each $P : D \to U_1$, compare the negative tests $P(d) \to \bot$ and $P(e) \to \bot$. With the encoded stable biconditional, define

$$E(d, e) := \Pi P : D \to U_1.\big(\neg_c P(d) \leftrightarrow_c \neg_c P(e)\big) : U_5.$$

Equality reflexivity, symmetry, and transitivity follow pointwise by identity, reversal, and composition of implications. Every typed $F : D \to D$ respects E: a test on $F(d)$ is pulled back to the test $u \mapsto P(F(u))$ on d. `Unary.equality_congruence_typed` implements that argument. It yields congruence for successor and both arguments of the auxiliary arithmetic operations.

Consequently the arithmetic predicates are

$$A^*(d, e, f) := E(\mathrm{Add}(d, e), f), \qquad M^*(d, e, f) := E(\mathrm{Mul}(d, e), f).$$

Their equality compatibility follows from operation congruence and the equivalence laws. Equality is thus a definable logical relation; it is not asserted to coincide with target beta conversion.

3.2 Successor-closed observation tests

A unary observation consists of the dependent telescope

$$\mathcal{U} = (X : U_1)(x_0 : X)(s : X \to X)(\delta : X \to U_4)(d_0 : \delta(x_0))(d_s : \Pi u : X.\delta(u) \to \delta(su)).$$

For $d : D$, its observation is $\mathrm{ev}_{\mathcal{U}}(d) = d\, X\, x_0\, s$. Define the positive predicate

$$G(d) := \Pi \mathcal{U}.\delta(\mathrm{ev}_{\mathcal{U}}(d)) : U_6.$$

For a test $P : X \to U_1$, its upward certificate says

$$\mathrm{Up}(P) := \Pi u : X.\delta(u) \to \neg_c P(u) \to \neg_c P(su).$$

Order transports a negative test through every certified observation:

$$O(d, e) := \Pi \mathcal{U}.\Pi P : X \to U_1.\, \mathrm{Up}(P) \to \neg_c P(\mathrm{ev}_{\mathcal{U}}(d)) \to \neg_c P(\mathrm{ev}_{\mathcal{U}}(e)) : U_6.$$

The observation quantifies only the displayed initial, successor, and upward certificates. Reflexivity and transitivity of O are derived by identity and composition; they are not extra

observation premises. Specializing equality to the pulled-back test $u \mapsto P(\mathrm{ev}_{\mathcal{U}}(u))$ proves $E(d,e) \to O(d,e)$. Symmetry of E and transitivity then give congruence of order.

The beta fold equations establish zero and successor closure of G. To derive `Unary.good_observed_typed`, use the rank-ten motive $u \mapsto J_9 \to G(u)$ in the `Good` certificate and evaluate at the canonical inhabitant of J_9. This explicit rank raising replaces an unavailable cumulativity step. Finally $G(d)$ supplies the guard required to derive $O(d, Sd)$ from the upward certificate. Repeating this step gives `Unary.ground_order_realized` for every external $i \leq j$.

4 Finite separation: turning a wrong comparison into Bottom

Theorem 4.1 (No backward numeral step)**.** *For every external n, there is an original inhabitant of $O(\nu_{n+1}, \nu_n) \to \bot$. The declaration is `no_back_realized` in the `Separation` module.*

Proof. Set $c = n + 1$ and $q = c + 1$. Use the selector carrier $F_q = U_1 \to \cdots \to U_1 \to U_1$ with selectors $e_0, \ldots, e_c$, zero e_0, and the capped successor table $s(e_i) =_\beta e_{\min(i+1,c)}$. The shared table-action and iteration theorems prove that the numeral ν_k evaluates to $e_{\min(k,c)}$ in this observation.

Its positive domain predicate is

$$\delta_q(u) := \Pi Q : F_q \to U_3.Q(e_0) \to \cdots \to Q(e_c) \to Q(u) : U_4.$$

Selector projection supplies the initial certificate. For successor closure, specialize an input certificate to $v \mapsto Q(sv)$ after opening the desired output predicate Q. Each resulting branch reduces to a selector branch already supplied by the continuation.

Take the separating test

$$P(u) := u \underbrace{\top \cdots \top}_{c \text{ entries}} \bot.$$

The upward law is proved by membership elimination at the rank-three motive $v \mapsto \neg_c P(v) \to \neg_c P(sv)$. At the cap it is identity. Below the cap its premise is $\top \to \bot$, which yields $\bot$ when applied to the known truth inhabitant; abstraction supplies the negative conclusion. These branches construct the required upward certificate.

Now specialize a purported $O(\nu_c, \nu_n)$ proof to this complete, closed observation. It gives $(\bot \to \bot) \to (\top \to \bot)$. Applying the two available identity proofs yields $\bot$. □

`Separation.ground_order_refutation` extends this result to every reversed ground comparison, using a forward comparison from $n + 1$ to the larger index. `Separation.R3_realized` refutes equality of any two distinct numerals by $E \Rightarrow O$ and, when necessary, symmetry of E. These finite observations have internal positive certificates; no parametricity principle claiming that all raw selectors are canonical is used. They are separating observations, not finite models of every source axiom.

5 Bounded cases and the quantified schemes

The key all-index theorem classifies an admitted presentation with a global order alternative:

$$\mathrm{Case}_n(d) := \bigvee_c \big(E(d,\nu_0),\dots,E(d,\nu_n),O(\nu_{n+1},d)\big) : U_8.$$

The last alternative is an actual global lower bound. Merely observing overflow in one finite algebra would not provide it.

Theorem 5.1 (Bounded cases)**.** *For every external cutoff n,* `Unary.bounded_case_realized` *constructs a closed original proof of* $\Pi d : D.\,\mathrm{Good}(d) \to \mathrm{Case}_n(d)$.

Proof. Use the positive rank-ten invariant

$$\mathrm{Inv}_n(d) := \Pi Z : U_8.(G(d) \to \mathrm{Case}_n(d) \to Z) \to Z.$$

At zero, supply its positive observation certificate and the reflexive equality branch. At successor, open the parent's continuation and eliminate the child invariant at that same Z. This supplies both $G(d)$ and the child case proof. Construct $G(Sd)$ from $G(d)$ independently. To construct the new negative case proof, open all branch continuations, so that the remaining answer type is $\bot$.

An equality branch with index $i < n$ moves to $i+1$ by successor congruence. At $i = n$, successor congruence and equality-to-order give $O(\nu_{n+1}, Sd)$. The already-large branch composes its lower bound with $O(d, Sd)$, whose positive guard is the retained $G(d)$. This proves successor closure of the invariant without extracting positive evidence from a negative disjunction.

Apply the original `Good` certificate to this exact rank-ten invariant. Project its case component by taking $Z = \mathrm{Case}_n(d)$ and the continuation $\lambda g\, c.c$. The source proves formation of the entire quantified case type at rank thirteen as well. □

The bounded-order scheme now follows in both directions. Under $O(d,\nu_n)$, a finite branch gives the desired finite equality disjunction. The large branch composes to $O(\nu_{n+1},\nu_n)$, contradicting separation. Conversely, an $i \le n$ equality branch gives $O(d,\nu_n)$ through equality-to-order and the ground comparison. `UnaryBounded.R4_realized` assembles the biconditional.

For comparability, a finite branch gives $O(d,\nu_n)$, while a large branch composed with $O(\nu_n,\nu_{n+1})$ gives $O(\nu_n,d)$. `UnaryAxioms.R5_realized` supplies the resulting disjunction. The first two graph schemes follow from the auxiliary arithmetic beta equations for arbitrary admitted third arguments. Numeral inequality is the separation result. `UnaryArithmetic.all_axioms_realized` and `RelationalAxioms.source_axiom_realized` connect these constructions to the literal source axiom syntax.

6 Preserving deductions and characterizing ground truth

The translation uses an environment from source variables to original PTS terms. `term_typed` and `term_admitted` prove that source terms remain in the admitted carrier when their environment entries do. Source renaming and substitution commute with translation; in particular, `instantiate_translation` identifies the translated universal instance with target substitution.

`RelationalEquality.formula_transport` extends atomic congruence to all formulas and `equality_replacement` realizes the source replacement rule. The important restriction is that

source variables range over admitted arguments. The proof does not assume or require that `Good` is invariant under equality between arbitrary raw presentations.

`RelationalDeduction.preservation` is proved by induction on the independent source derivation. Its open-context premises say that environment entries are typed and admitted and that translated source hypotheses are inhabited. Universal introduction extends the target context with both a carrier variable and its domain proof. Universal elimination uses term admission and the substitution identity. Classical elimination uses formula stability, and equality replacement uses the transport theorem. The closed specialization is `RelationalDeduction.sentence_preservation`.

For ground truth, a closed source term has a natural-number value and translates to the corresponding unary numeral up to beta conversion. Equality and order are then handled by reflexivity, forward comparisons, and separation. Addition and multiplication atoms also use the two arithmetic beta equations. `DirectRRelationalGround.v` proves each true and false case, and `Theorem4.ground_correctness` packages all four atom forms. These are proofs and internal refutations for every ground atom, not a finite regression sample and not reflection of arbitrary sentences.

7 Why the interpretation is not faithful

Consider $\rho := \forall x.(x \leq x)$. Its translation is inhabited by `Unary.admitted_order_reflexivity`: after opening the domain and observation binders, the proof returns its negative-test argument unchanged. Order is reflexive even on raw presentations, so the domain proof is not needed by this term.

For source non-derivability, the shared module `SourceCountermodel` uses $\mathbb{N} \sqcup \{\omega\}$. Successor fixes ω; the arithmetic graphs have the usual values on natural inputs and value zero when an input is ω. Order is usual on naturals, each natural is below ω, and ω is below no point. Thus every standard numeral has exactly its expected finite downset, and every point is comparable with a standard numeral, but $\omega \not\leq \omega$.

`RelationalCountermodel` defines evaluation and satisfaction for the independent source syntax, proves each axiom, and proves deduction soundness. Satisfaction is stable by structural induction: atomic relations in this model are decidable, and implication and universal quantification preserve stability. This justifies the classical source rule constructively, without a native excluded-middle axiom. `order_reflexivity_underivable` therefore gives

$$\widetilde{R} \nvdash \rho, \qquad \text{while} \qquad \vdash \mathrm{orderRefl} : \rho^*.$$

`Theorem4.nonfaithfulness` packages this witness. The countermodel is a model of the source arithmetic, not a set-theoretic model of the PTS. Its exceptional point also explains why nonfaithfulness does not conflict with the complete ground proof/refutation construction.

8 Reading map, verification, and limits

`DirectRUnary.v` contains the fixed terms, rank and typing certificates, basic equality/order laws, and bounded-case invariant. The evaluation, separation, bounded, arithmetic, and unary-axiom modules build the five schemes. The `DirectRRelational*` modules then supply independent syntax, structural translation, formula transport, deduction preservation, ground results, and countermodel soundness. `DirectRTheorem4.v` is the public endpoint. Shared logic, selectors, and certificate construction are documented by the parent development's exposition.

The central ranks do not depend on an axiom's numeral indices or a finite observation's size:

Construction	Rank
Unary carrier D	3
Finite domain certificate δ_q	4
Equality and arithmetic graph atoms	5
Positive observations G and order O	6
Bounded cases	8
Positive invariant	10
Admitted domain predicate	12

These are checked sufficient ranks; no optimality theorem is claimed. The construction provides neither induction for all raw carrier terms nor reflection of arbitrary source derivability. Internal refutations have type $A \to \bot$; external uninhabitability of $\bot$ is a separate metatheoretic assertion, not a conjunct of Theorem 4.

The library name remains `Interpretation_of_R.relational`. To inspect its public endpoint in Rocq:

```
From Interpretation_of_R.relational Require Import DirectRTheorem4.
Check Theorem4.theorem4.
Print Assumptions Theorem4.theorem4.
```

With the pinned environment described in `docs/DirectR-CI.md`, build from the repository root:

```
dune build @theories/Interpretation_of_R/relational/audit
```

The alias includes the unary and relational audit modules and depends only on this interpretation, the shared library, and the original kernel. It uses OCaml 4.14.2, Dune 3.23.1, Rocq 9.2.0, Stdlib 9.1.0, and Equations 1.3.2+9.2. Use a fresh absolute Dune build directory when fresh assumption output is required.

The retained 14 October layout audit prints the exact public theorem above followed by "Closed under the global context", and records successful fresh compilation and recursive kernel checking. Its `COMMANDS.md` and `types.log` are under `docs/direct-r-handoff/verification/theories-layout-2026-10-14`. The loaded-context inventory's unchanged upstream functional extensionality is distinct from the theorem's closed transitive assumption report. More recent verification is recorded in the live handoff. `bash tools/check-local-ci.sh` runs the complete staged local audit, including pristine upstream; compiling this exposition alone is not a fresh Rocq audit.